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Effects of different microbial pretreatments on the anaerobic digestion of giant grass under anaerobic and microaerobic conditions

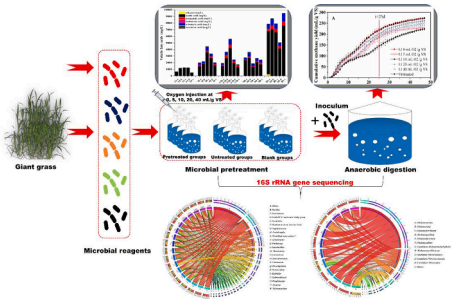
Ligong Wang, Caiyan Liu, Baocheng Wei, Chao Song, Fanfan Cai, Guangqing Liu, Chang Chen*

College of Chemical Engineering, Beijing University of Chemical Technology, Beijing 100029, China

HIGHLIGHTS

- Effects of loaded O_2 in microbial pretreatment on anaerobic digestion were studied.
- Higher yield of volatile fatty acids after microaerobic pretreatment was detected.
- Anaerobic pretreatment by straw composting inoculant had the highest methane yield.
- Microbial responses to O_2 loading varied significantly with microbial consortium.

GRAPHICAL ABSTRACT



ARTICLE INFO

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Methane yield
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ABSTRACT

Microbial pretreatment to lignocellulosic biomass for anaerobic digestion (AD) has achieved increased attention; however, the low efficiency and unclear mechanism of oxygen parameter affecting this process performance limit its practical application. In this study, five readily available microbial consortia were developed to analyze the influences of various oxygen concentrations during pretreatment process upon methane conversion efficiency and microbiota within AD of giant grass. Results found that anaerobic pretreatment by liquid or straw composting inoculant, along with microaerobic pretreatment by cow manure at 10 mL/g VS oxygen concentration, obtained 23.1%, 24.4%, and 16.0% higher methane yields (275.3, 279.8, and 265.3 mL/g VS) than corresponding untreated group, respectively. Microbial community analyses showed that microbial responses to oxygen varied significantly with microbial consortium, which consequently caused different AD performances. The findings will enrich theoretical knowledge of microbial pretreatment and provide a technological guidance for efficient utilization of giant grass and other lignocellulosic biomasses.

1. Introduction

With a significant focus on the revolution of clean energy, renewable energy has increasingly been considered as a main selling point for promoting this transition. Renewable methane obtained by anaerobic digestion (AD) can be a potential source of heat or electricity with negligible environmental issues, and exhibits promise in alleviating the current dependence on energy generated from fossil fuels. In addition,

* Corresponding author at: A505 Zonghe Building, Beijing University of Chemical Technology, 15 North 3rd Ring East Road, Beijing 100029, China.
E-mail address: chenchang@mail.buct.edu.cn (C. Chen).

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AD byproducts, such as biogas slurry (BS), can also be served as fertilizer (Peng et al., 2019). For these advantages, AD has attracted increasing attention and stimulated the rapid growth of AD enterprises worldwide. Nevertheless, a heavy dependence on feedstocks with seasonal characteristics, such as corn stalk, wheat stalk, and cotton stalk, has become a serious challenge to the stability of continuously operating AD reactors and even to the sustainable advancement of the AD industry (Zheng et al., 2015). This increasing awareness of the lack of AD feedstocks has inspired research on new renewable materials for a wide range of applications.

Giant grass (*Pennisetum giganteum* z.x.lin), an available energy crop with high productivity of over 500 tons/ha, is a perennial, renewable, and highly surviving herb mainly used to remediate environmental disturbances and produce biomass energy (Kang et al., 2016). Similar to conventional AD feedstocks, abundant fermentable organic matter, including cellulose and hemicellulose, is contained in giant grass and may allow this grass to serve as an attractive energy crop candidate (Liu et al., 2020). Nevertheless, the natural recalcitrance and high crystallinity of lignocellulose make hydrolytic enzymes with limited access to fermentable organic matter (Barua et al., 2018; Collard & Blin, 2014). As a result, current AD processes are usually designed and operated based on the inclusion of a pretreatment step, which results in acceleration of hydrolysis or solubilization of lignocellulose components, followed by a reliable and continuous acidogenesis and methanogenesis period (Barua et al., 2018). Pretreatments usually involve three methods, i.e., physical, chemical, and biological technology, in which microbial pretreatment may represent a more attractive option because of the energy-saving and eco-friendly characteristics (Wagner et al., 2013). Despite these advantages, microbial reagents used for pretreatment, such as *Phanerochaete chrysosporium*, generally need to be cultivated for a long time (approximately 20 days) under an oxygen supply (Zabed et al., 2019), which conflicts with the strict anaerobic condition of the subsequent AD process. As such, in addition to the AD digester, another reactor must be designed for microbial pretreatment, which increases the burden on the investment side and presents an important barrier for practical application. Therefore, alternative microbial pretreatment methods with short pretreatment period and cost-effective characteristic are urgently needed.

Whether oxygen should be supplied in the microbial pretreatment process is highly controversial. For example, Amin et al. injected 0, 6, 12, and 24 mL/g VS oxygen in the microbial pretreatment process of rice stalk and concluded that an oxygen supply of 12 mL/g VS was an appropriate stimulator of methane generation (Amin et al., 2021b). However, insignificant effects or even decreases in methane production have also been observed when oxygen was supplied to the microbial pretreatment process (Díaz et al., 2011; Lim & Wang, 2013; Okuda et al., 2007). Evidently, microbial pretreatment under anaerobic and microaerobic conditions has varied effects on AD performance; however, available explanations are extremely limited thus far. Therefore, it is necessary to perform comprehensive and systematic research to clarify the mechanism of microbial pretreatment under anaerobic and microaerobic conditions.

This study selected five kinds of microbial reagents from various environments, namely, liquid inoculant (LI), straw composting inoculant (SI), cow manure (CM), sheep dung (SD), and BS, all of which are readily available and contain different microorganisms for lignocellulose hydrolysis, to pretreat giant grass to analyze the influences of varied oxygen concentrations upon the methane conversion efficiency and microbiota. Moreover, both the pretreatment and AD were performed in one reactor, which can decrease the invested cost and increase the operating convenience. Overall, this study was performed to (1) develop pretreatment strategies with short pretreatment period and cost-effective characteristic by utilizing different microbial consortia to enhance methane production, (2) identify the effects of microbial pretreatment under anaerobic and microaerobic conditions upon volatile fatty acids (VFAs) variations, kinetic characteristics of methane generation, and lignocellulosic structure alterations, and (3) determine microbial responses to oxygen from the pretreatment process to the AD period to identify the mechanism involved in microbial pretreatment under anaerobic and microaerobic conditions.

2. Materials and methods

2.1. Feedstocks and inoculum

Giant grass was purchased on a local farm, situated in Hebei Province within China, and initially processed using a chopper to a length range of 1–2 cm and subsequently milled to cross a 0.5 cm sieve before usage. LI and SI are commercially applicable for stalk compost. CM and SD were kindly provided from a farm within Shunyi District of Beijing and using deionized water so that these materials could be evenly distributed. BS and inoculum have been fully described previously (Ning et al., 2018). Table 1 presents the physical–chemical characteristics of the abovementioned materials.

2.2. Microbial pretreatment and anaerobic digestion

Glass bottles were developed into pretreatment reactors (total volume = 500 mL, effective volume = 100 mL). First, 2 g/L SI, CM, SD, and BS and 10 mL of LI were applied as microbial consortia to pretreat 10 g/L giant grass on a volatile solids (VS) basis, and then deionized water was added to the reactor until an effective volume of 100 mL was reached. The air inside the pretreatment reactor was initially removed with the help of a nitrogen purge for each pretreatment reactor, followed by a quick sealing operation using rubber stoppers. Later, oxygen was separately injected at concentrations of 0, 5, 10, 20, and 40 mL/g VS (on a VS basis) to create anaerobic and microaerobic conditions for the respective pretreatment groups and then preserved for 5 days at 37 °C. During this period, manual mixing of the reactor once every day was required for even blending, and the oxygen concentration in the headspace was monitored.

After the oxygen was completely consumed, the inoculum was added

Table 1
Physical-chemical properties of feedstocks and inoculum.

Parameters	Giant grass	LI	SI	CM	SD	BS
TS(%)[a]	12.47 ± 0.42	1.55 ± 0.01	86.37 ± 0.03	28.56 ± 1.07	15.91 ± 0.28	6.95 ± 0.49
VS(%)[a]	10.60 ± 0.36	1.20 ± 0.01	22.18 ± 0.55	11.77 ± 0.25	13.87 ± 0.40	3.98 ± 0.61
Ash(%)[a]	1.87 ± 0.03	0.35 ± 0.01	64.19 ± 0.46	16.79 ± 0.56	2.04 ± 0.34	2.97 ± 0.32
VS/TS(%)	84.99 ± 0.13	77.42 ± 0.02	25.68 ± 0.57	41.20 ± 0.43	87.18 ± 0.62	57.27 ± 0.91
Cellulose(%)[b]	33.94 ± 0.75	NA	NA	9.42 ± 0.80	18.97 ± 0.84	NA
Hemicellulose(%)[b]	27.21 ± 0.36	NA	NA	8.92 ± 0.13	27.11 ± 0.70	NA
Lignin(%)[b]	4.92 ± 0.15	NA	NA	10.70 ± 0.72	6.61 ± 0.16	NA
C(%)[b]	39.44 ± 0.79	NA	8.84 ± 0.41	19.81 ± 0.84	35.81 ± 0.92	30.31 ± 0.17
H(%)[b]	4.14 ± 0.21	NA	1.03 ± 0.03	2.44 ± 0.20	5.48 ± 0.17	4.05 ± 0.05

NA: Not available; [a] On the standard of the total weight; [b] On the standard of the TS.

to each reactor for AD at 10 g VS/L to ensure a favorable biodegradation rate (Amin et al., 2021b). The effective volume was adjusted to 200 mL with deionized water. To exclude the background methane yield from the pretreatment reagent for an accurate experimental value, control groups that included respective microbial consortia and inoculum were set for comparison. Meanwhile, untreated groups that included raw giant grass, inoculum, and respective microbial consortia were also set. After an adjustment of the initial pH to 7.5 with a NaOH solution, all AD reactors were flushed with nitrogen for 3–5 min to remove air for an anaerobic condition, and then subjected to 47 days of cultivation at a temperature of 37 °C. The reactor was shaken manually twice per day to ensure full contact of individual components. The daily biogas accumulation was computed based on a previously described method (Ning et al., 2018).

2.3. Analysis methodology

The measurement to quantify the values of total solids (TS) and VS followed previously reported procedures (Ning et al., 2018), while microelement (C and H) qualification was fulfilled with Germany's Vario EL CUBE element analyzer. The lignocellulosic components were characterized by means of the Van Soest procedure (Van Soest et al., 1991). The whole procedure for determining VFAs concentrations and compositions was accomplished by means of an Agilent 7890A gas chromatograph manufactured in USA. Quantification of oxygen during the pretreatment process and methane in the biogas was performed with an Agilent 7890B gas chromatograph (USA), and a relevant analytical column (Agilent Hayesep Q). Helium at 34.5 kPa pressure was used as the carrier gas during the operation. The biodiversity and structural analyses of microbial communities are detailed in the Supplementary Material. Origin software, single-factor analysis of variance, and circos online service have been used for the data analyses.

2.4. Kinetic and microstructure analysis

To further comprehensively assess the AD performance of giant grass, both a modified Gompertz model and a first-order dynamic model were applied to the accumulated methane production data obtained over 47 days, as described previously (Ning et al., 2018).

The lignocellulosic components and structural changes in giant grass with or without pretreatment were primarily determined based on the Van Soest method, X-ray diffraction (XRD), and Fourier transform infrared (FTIR) spectroscopy. The XRD system (Bruker, Germany) is equipped with a Cu Kα target at a scanning rate of 5.0°/min, with variations between 5° and 90°, and the FTIR system (Thermo Fisher Nicolet 6700, USA) is equipped with a DLATGS detector operating at scanning wavelengths in the range of 400 cm^{-1} to 4000 cm^{-1}.

3. Results and discussion

3.1. Effects of variant oxygen concentrations in the pretreatment process upon the yields of volatile fatty acids

VFAs, mainly acetic acid, propionic acid, butyric acid, valeric acid, etc., are well known as the primary products of the acidogenesis phase and the key carbon sources for yielding methane (Chen et al., 2012). Here, the concentrations and main components of VFAs after different microbial pretreatments at various oxygen concentrations were detected, as displayed in Fig. 1. The VFAs concentrations were found at various levels among all pretreatment groups, with a higher VFAs level observed for the SI, CM, SD, and BS pretreatments than the LI pretreatment. This result may be related to the effect of innate differences in the composition and microbial species within the pretreatment agents. Moreover, when oxygen was supplied to the pretreatment process, the likelihood of hydrolysis products transforming into VFAs was generally enhanced, which was similar with prior observations (Eryildiz and

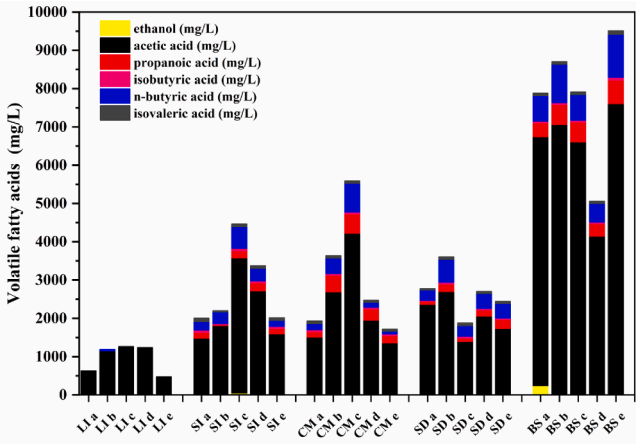


Fig. 1. Volatile fatty acids (VFAs) concentrations and compositions after microbial pretreatment at different oxygen concentrations of (a) 0 mL/g VS, (b) 5 mL/g VS, (c) 10 mL/g VS, (d) 20 mL/g VS, and (e) 40 mL/g VS.

LukitawesaTaherzadeh, 2020; Khan et al., 2016).

The VFAs concentrations from LI pretreatment reached a peak (1263.3 mg/L) at an oxygen concentration of 10 mL/g VS, and primarily corresponded to acetic acid, which was quite different from the results of other pretreatment groups. The VFAs concentrations of the SI pretreatment increased from 2001.4 to 4465.9 mg/L when the oxygen concentration ranged from 0 to 10 mL/g VS and subsequently decreased but remained higher (2017.4–3374.7 mg/L) than that under anaerobic condition. Among the compositions, acetic acid was expected to take the dominant position in a range from 1482 to 3533.8 mg/L. When the oxygen concentration was enhanced to 10 mL/g VS, the yields of acetic acid and butyric acid received maximum values of 3533.8 and 622.0 mg/L, respectively. At oxygen concentrations of 10 and 20 mL/g VS, additional ethanol products were detected at concentrations of 45.2 and 15.1 mg/L, respectively. The VFAs concentrations during CM pretreatment at oxygen concentrations of 0, 5, 10, 20, and 40 mL/g VS were 1932.8, 3636.5, 5590.3, 2471.1, and 1718.0 mg/L, respectively. The production of VFAs started to increase after oxygen addition during the pretreatment process and achieved the highest value at 10 mL/g VS of oxygen concentration, corresponding to 2.89-fold than that of the anaerobic pretreatment. This result might be associated with the better microbial growth of the CM reagent under microaerobic conditions (Jackson et al., 1970). While, the suitable oxygen concentration for the SD reagent was 5 mL/g VS, resulting in a VFAs production of 3606.7 mg/L, in which acetic acid accounted for 2699.5 mg/L. In comparison with other pretreatments, BS at any oxygen concentration produced higher VFAs production, which changed from 4734.6 to 8388.4 mg/L. The high output of VFAs might be a consequence of the abundant bacteria related to the hydrolysis and acidogenesis in BS.

3.2. Cumulative methane yield analysis

Effects of oxygen introduced at different loadings during the microbial pretreatment process on cumulative methane yield (CMY) are indicated in Fig. 2. The experimental methane yield (EMY), an indication of the maximum CMY, has been applied as an indicator for AD performance evaluation. The anaerobic pretreatment by LI resulted in a higher rate of methane accumulation, which reached an EMY of 275.3 mL/g VS (Fig. 2A). Oxygen supplied at 5 mL/g VS to the LI pretreatment process led to a decrease in the EMY to 230.6 mL/g VS, and an increase in oxygen from 10 to 40 mL/g VS led the EMY to vary from 249.4 to 272.6 mL/g VS. Anaerobic pretreatment by LI produced the highest EMY increase (23.1%) in relation to the untreated group (223.6 mL/g VS). In the SI pretreatment groups (Fig. 2B), and it reached 279.8 mL/g VS without oxygen in the SI pretreatment process with an increase of 24.4%

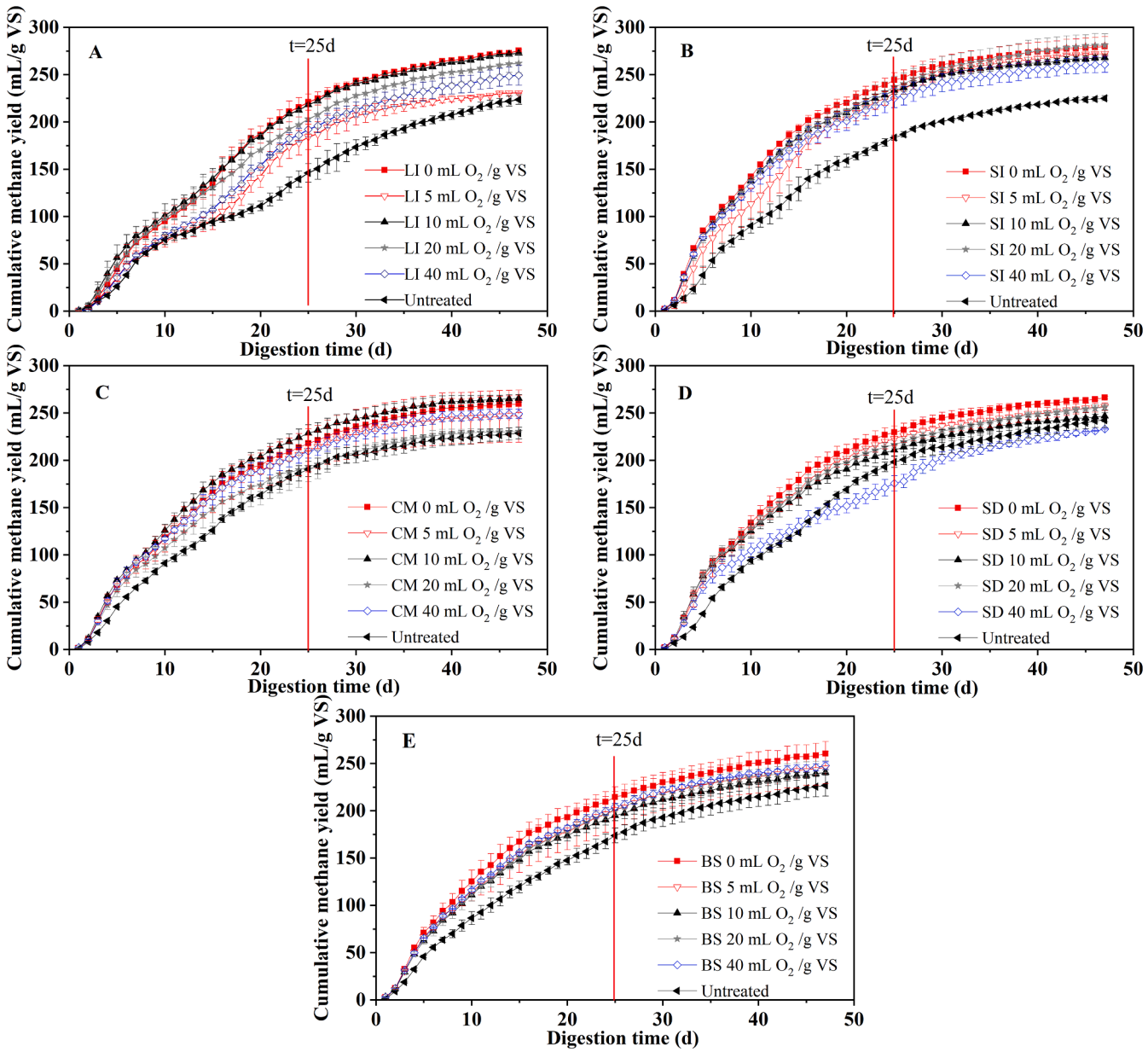


Fig. 2. Cumulative methane yield from giant grass after (A) LI, (B) SI, (C) CM, (D) SD, and (E) BS pretreatment at different oxygen concentrations.

than untreated group (225.0 mL/g VS), but decreased after injecting either 5 mL/g VS oxygen or 10 mL/g VS oxygen. Thereafter, the EMY peaked at 281.3 mL/g VS under continuous increases in the oxygen concentration to 20 mL/g VS, with the highest increase of 25.0% relative to the untreated group, and then the value decreased to 260.8 mL/g VS at 40 mL/g VS oxygen. Notably, the differences in EMY between SI pretreatments under anaerobic conditions and 20 mL/g VS oxygen concentration were limited. In other words, oxygen was not recommended for the SI pretreatment process because of the extremely limited enhancement in EMY. The EMYs of the CM pretreatment at 0, 5, 10, 20, and 40 mL/g VS oxygen concentrations were 259.5, 246.6, 265.3, 230.5, and 249.0 mL/g VS, respectively, which represented increases of 0.8–16.0% relative to the untreated group (228.6 mL/g VS). Methane production showed the highest increase at an oxygen concentration of 10 mL/g VS, probably as a result of the highest VFAs concentration achieved during pretreatment process under this condition. Regarding the SD pretreatment group, the maximum EMY of 266.3 mL/g VS occurred under anaerobic conditions, which was a 9.9% increase compared with that of the untreated group (242.2 mL/g VS). Moreover, giant grass pretreated with SD at 40 mL/g VS oxygen unexpectedly showed an inferior methane output of 233.1 mL/g VS compared with untreated grass. Consistent with the SD pretreatment groups, the application of a microaerobic strategy for the BS pretreatment process also caused a reduction in EMY. In the BS pretreatment group, anaerobic condition achieved the highest EMY of 260.2 mL/g VS, which represented an improvement of 14.8% in relation to untreated group (226.7 mL/g VS). This were likely associated with the relatively higher VFAs concentrations (more than 5000 mg/L) during the anaerobic pretreatment process.

In short, higher EMYs were commonly determined for the LI, SI, and CM pretreatment groups at optimum conditions, which represented increases of 23.1, 24.4, and 16.0% than the untreated groups, respectively. Fu et al. analyzed the performance of microaerobic pretreatment under thermophilic condition and attained a maximum increase of 16.2% at an oxygen concentration of 5 mL/g VS with respect to untreated straw (Fu et al., 2015). Similarly, Xu et al. performed a biological pretreatment using a pure bacterial system with or without introducing oxygen for AD of corn straw, and received the maximum EMY of 270.8 mL/g VS at an

operating oxygen concentration of 5 mL/g VS, along with a 17.4% increase relative to the untreated system (Xu et al., 2018). Comparatively, the enhanced methane production observed in these previous studies could be achieved using CM at suitable oxygen concentration in the present work, not to mention LI and SI pretreatment. Furthermore, using CM could reduce pollution by transforming waste material into a useful end product while also reducing the pretreatment cost. Therefore, future AD applications should focus on these readily available pretreatment agents. One more thing, after an AD run of 25 days, the CMY from all groups amounted to 80.2–86.2% of the final EMY. This implied that if the digestion time were designed as 25 days instead of 47 days, the size of future AD reactor could be decreased by 46.8% by sacrificing only 13.8–19.8% of methane production.

3.3. Kinetic and microstructure analysis

The understanding of kinetic characteristics is important for future modifications of technical procedures (Xu et al., 2015). Thus, the classical modified Gompertz and first-order dynamic models are implemented. As pointed out in the resulting Table 2, the reliability of further analysis using the simulated kinetic parameters was confirmed because the correlation coefficients (R^2) of these two models were greater than 0.98. Consistent with the EMY data, the simulated maximum CMY (B_0) from both models varied significantly with changes in either the pretreatment agent or oxygen loading. Commonly used to indicate the lag phase, λ tended to decrease after the SD, SI, and BS pretreatments and increased to different extents after the CM and LI pretreatments, as shown in the illustration in Fig. 2. The hydrolysis rate (k) was expected to increase in the pretreatment groups compared with the untreated samples, and the highest hydrolysis rate generally corresponded to the highest EMY for each pretreatment group.

To better understand the characteristic changes in giant grass before and after oxygen loading in the pretreatment process, XRD and FTIR techniques were used to provide valuable information, such as the crystalline degree of cellulose and the chemical composition. The LI, SI, and CM groups had high EMY values and were selected for the XRD and FTIR analyses. The patterns included two peaks (see Supplementary Material) at $2\theta = 16^\circ$ and $2\theta = 22^\circ$ for cellulose I crystals and $2\theta = 18^\circ$ for amorphous substances. Most peaks at $2\theta = 22^\circ$ for the pretreated samples decreased to different extents, thus indicating that the crystalline degree of cellulose was transforming to some extent. Amorphous cellulose is more easily utilized, and the intensity approximately at $2\theta = 18^\circ$ was detected for the pretreated samples; however, it was not observed in the untreated samples. This result was indicating that pretreatment certainly reduce the crystalline degree of cellulose, as shown in the Supplementary Material. The peaks observed in the FTIR spectra could be explained as follows (see Supplementary Material): 1730 cm^{-1} for the C = O vibration, 1515 cm^{-1} for the benzene framework, and 900 cm^{-1} for the O–H stretching vibration (Correia et al., 2013; Ding et al., 2012; Faulon et al., 1994; Pandey & Pitman, 2004). Peaks at 1730 cm^{-1} correlated well with the hemicellulose composition (Ding et al., 2012; Pandey & Pitman, 2004), and obvious alterations in this peak were not found between anaerobic- and microaerobic-pretreated giant grass, although the peak decreased compared to that of the untreated giant grass. Hemicellulose has been widely studied as a readily decomposing component; thus, its dissolution after pretreatment could be helpful for VFAs and EMY production, as previously observed. On the other hand, peaks at 1515 cm^{-1} were the characteristic of lignin (Pandey & Pitman, 2004). A decrease in the height of this peak was observed for all pretreated giant grass relative to the untreated sample. Lignin could greatly hinder cellulose degradation; thus, its removal has attracted extensive attention. Therefore, all pretreatments were expected to facilitate

Table 2
Kinetics parameters of model fitting.

Pretreatment	Oxygen concentration (mL/g VS)	Modified Gompertz model				First order dynamics model		
		B_0	μ_m	λ	R^2	B_0	k	R^2
LI	Untreated	250.2	6.2	0.6	0.987	408.9	0.018	0.992
	0	277.8	10.8	1.7	0.996	353.6	0.036	0.985
	5	242.9	8.6	2.3	0.991	342.6	0.027	0.975
	10	276.6	10.3	0.7	0.994	336.4	0.039	0.991
	20	268.6	9.3	0.6	0.994	334.2	0.036	0.993
	40	257.3	9.0	2.1	0.995	364.8	0.027	0.984
SI	Untreated	224.3	9.3	1.1	0.994	267.2	0.043	0.989
	0	274.1	13.6	−0.2	0.990	295.1	0.068	0.994
	5	272.6	11.9	0.7	0.994	310.5	0.051	0.990
	10	263.0	12.9	−0.2	0.990	284.1	0.067	0.994
	20	276.4	12.4	−0.4	0.989	302.5	0.060	0.995
	40	256.0	12.1	−0.4	0.990	276.4	0.066	0.995
CM	Untreated	230.5	9.4	0.6	0.987	272.9	0.044	0.988
	0	277.8	10.8	1.7	0.996	286.1	0.056	0.955
	5	242.9	8.6	2.3	0.991	269.7	0.060	0.994
	10	276.6	10.3	0.7	0.994	288.3	0.061	0.994
	20	268.6	9.3	0.6	0.994	252.3	0.058	0.996
	40	257.3	9.0	2.1	0.995	272.0	0.059	0.995
SD	Untreated	241.7	9.7	1.3	0.994	295.4	0.040	0.986
	0	260.3	12.4	−0.4	0.990	280.7	0.067	0.994
	5	251.3	12.0	−0.3	0.991	272.1	0.065	0.993
	10	242.5	10.8	−1.0	0.987	259.6	0.066	0.995
	20	250.4	11.1	−1.0	0.987	269.0	0.065	0.996
	40	239.2	7.4	−2.6	0.981	261.7	0.047	0.992
BS	Untreated	229.4	7.8	−0.2	0.993	276.1	0.038	0.997
	0	252.9	10.8	−0.9	0.987	275.3	0.060	0.996
	5	242.5	9.6	−1.1	0.988	265.4	0.056	0.997
	10	236.0	9.4	−1.0	0.989	259.4	0.056	0.997
	20	243.1	9.8	−0.8	0.990	268.7	0.054	0.997
	40	244.0	10.0	−0.9	0.990	267.3	0.057	0.997

B_0: simulated maximum cumulative methane yield (mL/gVS), μ_m: simulated maximum methane production rate (mL/gVS), λ: lag phase time (d), R^2: correlation coefficient.

cellulose decomposition for methane production. More specifically, the CM and SI pretreatments under anaerobic and microaerobic conditions did not show a significant difference in the peak at 1515 cm^{-1}. However, an increased height of this peak was observed under microaerobic conditions compared with anaerobic conditions for the LI pretreatment. Consequently, anaerobic LI pretreatment could favor methane production in giant grass more than microaerobic conditions, as displayed in Fig. 2. The peak at approximately 900 cm^{-1} came from the stretching vibration of O–H, a characteristic absorption band indicating cellulose. All pretreatments caused a decrease in the height, indicating that cellulose had been partially degraded during the pretreatment process, which was consistent with the information in Supplementary Material.

3.4. Biodiversity and microbial communities analyses

3.4.1. Biodiversity and species abundance distribution

LI pretreatment groups at 0 and 10 mL/g VS, SI pretreatment groups at 0 and 20 mL/g VS, and CM pretreatment groups at 0 and 10 mL/g VS were selected for microbial communities' analyses due to the relatively good EMY performance. Various indicators, including the OTU number, Shannon index, Simpson index, ACE, and Chao 1 estimator, can represent the level of microbial diversity and richness, and were compared (see Supplementary Material). A coverage value over 0.98 indicated reliability for further analyses. It was suggested that bacteria did generally occur in higher Shannon index, ACE, and Chao 1 estimator than archaea, and therefore in predominant microbial community from pretreatment until AD process. The CM group had higher species diversity and richness during the pretreatment process and higher Shannon index and ACE estimator values for bacteria, which were likely the inherently abundant microorganisms in CM. Additionally, the high values observed for CM might explain the higher VFA concentrations shown in Fig. 1. Except for the ACE estimator value obtained in the LI and CM pretreatment processes, the Shannon index and ACE estimator of bacteria from the remaining experimental groups showed enhancements with the introduction of oxygen in the pretreatment process. After AD, the Shannon index and ACE estimator of bacteria showed different degrees of increase among the different pretreatment processes, which was likely caused by the appearance of new bacterial species associated with AD. Archaea play a critical role in the final methane conversion efficiency. Compared to anaerobic conditions, the ACE and Chao 1 estimator values of the digestate from microaerobic CM-pretreated giant grass increased, while those from microaerobic LI-pretreated giant grass did not show significant differences, which may explain the higher EMY resulted from microaerobic pretreatment by CM and anaerobic pretreatment by LI.

3.4.2. Bacterial analysis on the general level

Bacterial relative abundance (RA) on the general level in the pretreatment agent, during the pretreatment process, and after AD was detected, as shown in Fig. 3. The dominant bacterium in the LI reagent was *Bacillus* (88.5% RA), which then decreased to 54.0% and 46.4% during the pretreatment under anaerobic and microaerobic condition, respectively. The RA of *Bacillus* finally decreased to 0.2% and 0.5% after the AD period, respectively. Generally, anaerobic conditions could enrich members of this genus more than microaerobic conditions in the pretreatment process. *Bacillus* is an aerobic or facultative anaerobe that can release many extracellular enzymes, such as cellulase, xylanase, amylase, and peroxidase (Carrillo-Reyes et al., 2016; Zabed et al., 2019). Cellulases are important enzymes that biocatalyze depolymerization of cellulosic biomass into fermentable sugars, while xylanase can convert hemicellulose into oligosaccharides and release easily degradable nutrients from lignocellulose biomass (Carrillo-Reyes et al., 2016; Zabed et al., 2019). Peroxidase is highly linked to lignin decomposition (Carrillo-Reyes et al., 2016). Thus, *Bacillus* had a strong relationship with

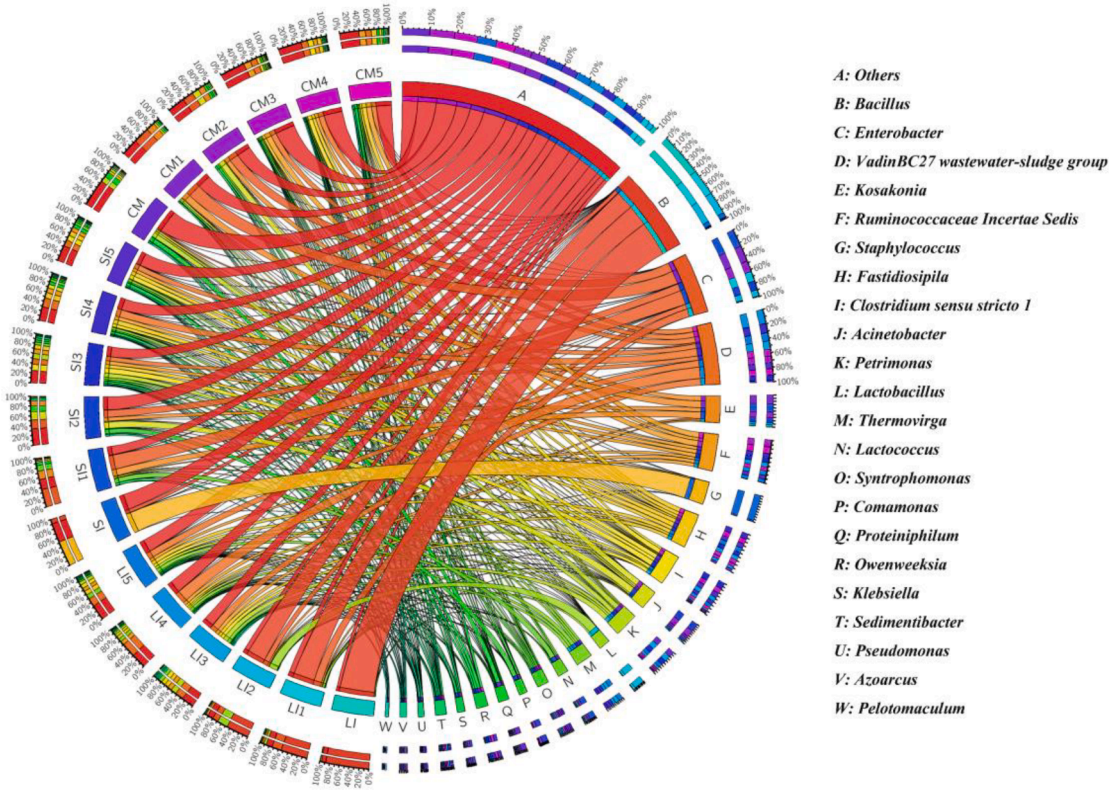


Fig. 3. Relative abundance of bacteria on general level. (1, 2): effluent from giant grass after microbial pretreatment under anaerobic and optimal microaerobic condition, respectively; (3, 4, 5): digestate from untreated, anaerobic microbial pretreated, optimal microaerobic microbial pretreated giant grass after AD, respectively.

cellulose and lignin decomposition, and an increase in RA under anaerobic conditions led to greater lignocellulose degradation compared with that under microaerobic conditions. This result could be reflected in the analyses of lignocellulose alteration (see Supplementary Material), where the hemicellulose content was generally released after the anaerobic pretreatment by LI and could be readily used in the subsequent AD period. Additionally, the RA of the second most dominant genus (*Enterobacter*) was significantly different under anaerobic conditions (19.7%) and microaerobic conditions (12.5%). After the AD period, this genus was scarcely detected. *Enterobacter*, as a facultative anaerobe, was reported to belong to the family of *Enterobacteriaceae*, and previous literature commonly applied anaerobic condition to promote the enrichment of this bacterium, which might certainly explain why *Enterobacter* has a higher RA under anaerobic condition than that under microaerobic condition (Maru et al., 2016). The function of *Enterobacter* includes transforming multiple carbon sources into acetic acid and H_2 (Maru et al., 2016), and has a potential beneficial effect in cooperation with methanogenic bacteria. These results indicated that applying anaerobic conditions could help to enrich *Enterobacter* during LI pretreatment process and thus might benefit subsequent AD performance. The third most dominant genus is *Lactobacillus*, which is strongly positively correlated with the conversion of a wide range of sugars into lactate (Hatti-Kaul et al., 2018). This genus was more prevalent under microaerobic conditions (20.9%) than under anaerobic conditions (4.8%). Moreover, the bacteria *Clostridium sensu stricto 1*, *Sedimentibacter*, *Ruminococcaceae incertae sedis*, *Petrimonas*, *VadinBC27*, *Pelotomaculum*, *Azoarcus*, *Syntrophomonas*, and *Thermovirga* did not show significant growth in the pretreatment procedure but began to increase in the AD process. These microbes have the potential to transform several sugars and organic acids as acetic acid, butyrate, and H_2 to enhance the subsequent methanogenesis rate (Costabile et al., 2017; Grabowski et al., 2005; Li et al., 2018; Morrison et al., 2016; Nordhoff et al., 2005; Xie et al., 2014). Notably, the RA of these genera was generally higher in the digestate from anaerobic pretreatment than that microaerobic pretreatment by LI, which possibly played an important role in the greater accumulation of methane for the LI pretreatment group under anaerobic condition (Fig. 2).

The dominant genus in the SI agent was *Staphylococcus* at a RA of 51.4%. During the microbial pretreatment process, this bacterium disappeared; *Enterobacter*, *Kosakonia*, and *Lactococcus* started to develop as the dominant genera, and showed greater RA values under anaerobic pretreatment conditions (36.0, 16.9, and 11.5%, respectively) than under the microaerobic pretreatment conditions (18.5, 10.3, and 1.6%, respectively). However, *Enterobacter*, *Kosakonia*, and *Lactococcus* were almost undetected after AD process. As previously mentioned, *Enterobacter* could benefit subsequent AD performance; therefore, the finding of a higher methane yield from the giant grass pretreated by anaerobic SI pretreatment was reasonable (Fig. 2). *Kosakonia* is a facultative anaerobe, and is effective in the fixation of nitrogen nutrients and the production of xylanase (Chin et al., 2017). Nitrogen fixation might benefit the growth of hydrolytic microorganisms, while the generation of xylanase will promote the conversion of fiber into oligosaccharides and the release of fermentable nutrients from lignocellulose biomass (Chin et al., 2017). For *Lactococcus*, it is a part of the group of lactic acid bacteria, and is likely to induce a significant promotion on the generation of lactic acid, which was a useful source of VFAs (Casalta & Montel, 2008). Therefore, the higher RA of *Enterobacter*, *Kosakonia*, and *Lactococcus* probably laid a foundation for the observed higher methane production of anaerobically pretreated giant grass by SI. After the AD process, the dominant genera of the SI pretreatment groups included *VadinBC27*, *Ruminococcaceae incertae sedis*, *Fastidiosipila*, and *Clostridium sensu stricto 1*. Comparably, the *VadinBC27*, *Ruminococcaceae incertae sedis*, and *Fastidiosipila* presented higher RA after the AD period of giant grass under anaerobic condition (18.8, 9.6, and 10.9%, respectively) compared with the microaerobic condition (12.7, 8.2, and 9.5%, respectively). The *VadinBC27* belonged to the *Rikenellaceae* family of the *Bacteroidetes* phylum and was identified as a major factor for methane production (Rautio et al., 2003). It was recognized as degrading amino acids in syntrophic association with hydrogenotrophic methanogens, and the 48.0% increase in the RA for anaerobic condition than microaerobic condition might also played an important role in enhancing methane production. *Ruminococcaceae incertae sedis* hydrolyzes and digests carbohydrates, such as cellulose and hemicellulose, to form short-chain fatty acids (Costabile et al., 2017), while *Fastidiosipila* is important for generating precursor for methane synthesis (Falsen et al., 2005). *Clostridium sensu stricto 1* is one of the most common genera in AD and generally acts on several types of sugars for preparing methanogenesis precursors, and it therefore has the added benefit of reducing the starting time for AD (Lanjekar et al., 2015). However, the significant genera including *Clostridium sensu stricto 1*, *Ruminococcaceae incertae sedis*, and *Fastidiosipila*, did not have significant differences between the microaerobic and anaerobic SI pretreatment process. Over the whole, the microaerobic pretreatment decreased the RA of *Enterobacter* and *Kosakonia* and negatively influenced the enrichment with respect to the *vadinBC27* in AD process. Therefore, oxygen loading was not recommended during the SI pretreatment process.

Bacterial genera such as *Acinetobacter*, *Proteiniphilum*, *Petrimonas*, *Fastidiosipila*, and *VadinBC27* are common in CM. *Acinetobacter*, *Proteiniphilum*, and *Petrimonas* are most likely available for the decomposition of lignin, protein, and hemicellulose, respectively (Iino et al., 2009; Iyer & Mahadevan, 2002; Wongfaed et al., 2020). Most bacteria from *Acinetobacter* were aerobic, therefore during CM2, the RA of *Acinetobacter* generally increased, with values increasing from 4.5 to 14.2% compared to that with the CM reagent. This finding indicated that oxygen in the CM pretreatment might play an important role in lignin degradation; therefore, VFAs concentrations would increase under microaerobic condition (Fig. 1). However, this genus was rarely detected after the AD period, which might be related to its aerobic characteristics. Changes in the RA of *Proteiniphilum* and *Petrimonas* differed relative to that of *Acinetobacter,* and tended to decreased to a negligible level across both the CM pretreatment process and AD period. *Fastidiosipila* has been shown to promote the accumulation of acetic and butyric acids, known as precursors for methane production. *Fastidiosipila* and *VadinBC27* tended to change similarly; initially, the RA was hardly detected in pretreatment process regardless of anaerobic or microaerobic condition, followed by an increase after AD period and the abundance under anaerobic and microaerobic condition did not vary significantly. *Clostridium sensu stricto 1* is important for converting multiple sugars into methanogenic precursors (Morrison et al., 2016; Xie et al., 2014). These bacteria were not observed in the CM, and the increase in the RA was not obvious in the pretreatment stage; although a significant increase occurred after the AD period, the RA under anaerobic and microaerobic condition was not significantly different. In general, a limited oxygen supply (10 mL/g VS) would achieve a high RA of the *Acinetobacter* during CM pretreatment process, and might alleviate the lignin resistance to lignocellulosic hydrolysis. Therefore, VFAs were efficiently produced during the microaerobic CM pretreatment process while considerably high EMY values were observed in the AD of giant grass under this condition.

3.4.3. Archaeal analysis on the general level

The distribution and RA values of archaea between the anaerobic and microaerobic conditions for the LI, SI, and CM groups are shown in Fig. 4. The prominent genera in this study were mainly *Methanospirillum* (52.7%), *Methanobacterium* (32.7%), *Methanosaeta* (12.0%), and *Methanosarcina* (1.75%) in the LI reagent; *Candidatus nitrososphaera* (31.7%), *Methanosarcina* (25.6%), *Methanoculleus* (20.6%), *Methanomassiliicoccus* (5.9%), *Methanobacterium* (5.5%), and *Methanosaeta* (4.8%) in the SI reagent; and *Methanobrevibacterium* (68.8%), *Methanosaeta* (18.7%), *Methanoculleus* (5.0%), *Methanosarcina* (3.7%), and *Candidatus methanomethylophilus* (1.7%) in the CM reagent. *Methanosaeta* functions via the acetoclastic pathway to produce methane (Amin et al., 2021a), while *Methanosarcina* follows acetoclastic, hydrogenotrophic, and

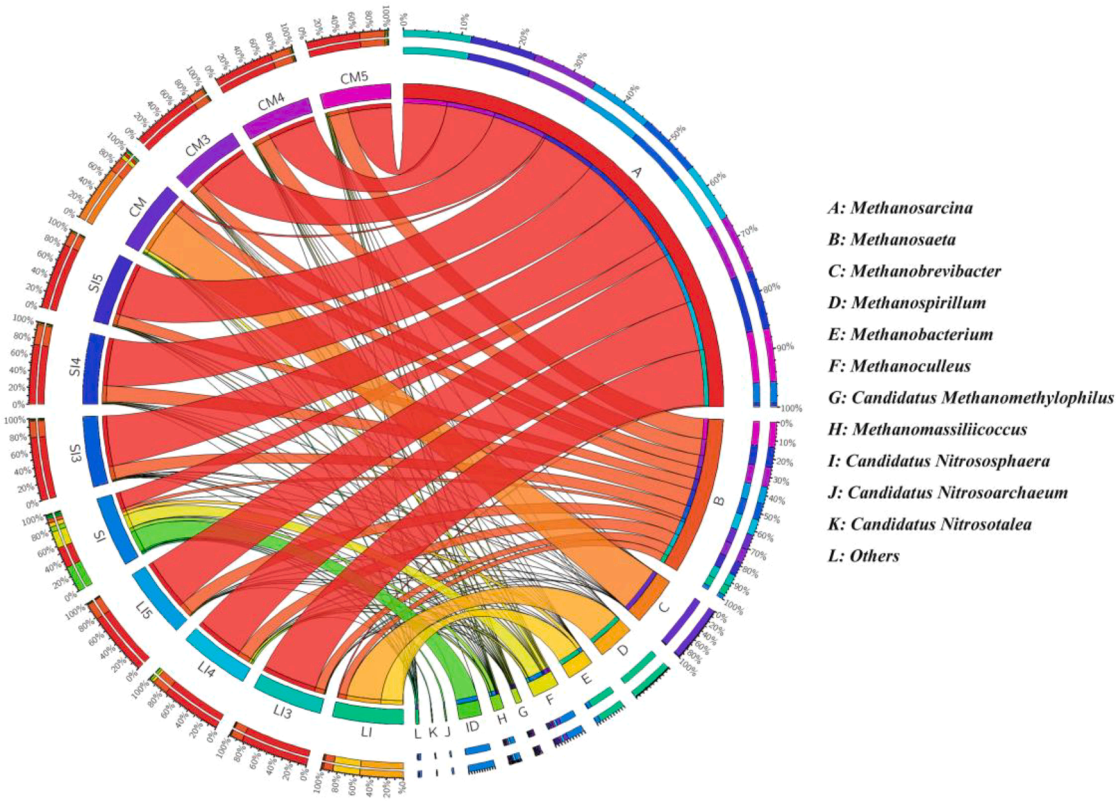


Fig. 4. Relative abundance of archaea on general level. (3, 4, 5): digestate from untreated, anaerobic microbial pretreated, optimal microaerobic microbial pretreated giant grass after AD, respectively.

methylotrophic pathways and could additionally capture H_2, CO_2, and methanol for methane production compared to *Methanosaeta* (Amin et al., 2021a; Pyzik et al., 2018; Shimizu et al., 2011). *Methanoculleus* has been found to function by hydrogenotrophic pathways and uses multiple materials, including H_2, CO_2, and formate, to form methane, and *Methanomassiliicoccus* is widely known to function via the methylotrophic pathway for methane production using methanol (Amin et al., 2021a). For LI pretreatment group, the RA of *Methanosarcina*, *Methanosaeta*, *Methanoculleus*, and *Methanomassiliicoccus* detected under anaerobic condition was 73.5, 19.8, 3.1, and 2.8%, while under microaerobic condition, it changed to 77.6, 20.5, 0.9, and 0.6%, respectively. Obviously, the response of microbial structures to oxygen varied with the microbial reagent, and thus, the microbial reagent might be a potential parameter influencing microorganisms within the digestion period.

4. Conclusions

This study identified commonly higher VFAs yield after microaerobic pretreatment, and 23.1%, 24.4%, and 16% increase in methane production than untreated respectively after anaerobic pretreatment by LI, anaerobic pretreatment by SI, and microaerobic pretreatment by CM at 10 mL/g VS oxygen concentration. Also, microbial response to oxygen depended on the microbial consortia. For example, anaerobic condition benefited the enrichment of *Bacillus* and *Enterobacter* during LI pretreatment process, which might promote the conversion of lignocellulose and enhanced AD performance. While, optimal oxygen could enrich lignin-degrading bacteria (*Acinetobacter*) in CM pretreatment process, thereby observing high EMY value under this condition.

CRediT authorship contribution statement

Ligong Wang: Formal analysis, Investigation, Writing-original draft, Software, Writing-review & editing. **Caiyan Liu:** Formal analysis, Investigation, Validation. **Baocheng Wei:** Resources. **Chao Song:** Resources. **Fanfan Cai:** Resources. **Guangqing Liu:** Methodology. **Chang Chen:** Conceptualization, Methodology, Funding acquisition, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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